Rust for Real: A Performance and WCET Comparison of Rust and C for Real-Time Systems

Appendix

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A) Full Data

Table 1: Benchmark Cycle Counts (kernel)

			Oz	Os	O0	O1	O2	O3
binary search	C	A	173	113	369	121	105	111
		M	123	76	297	97	79	80
	Rust	A	330	380	765	296	287	287
		M	144	146	588	140	138	138
bitcount	C	A	25794	24298	1.38325e+05	24328	19664	19289
		M	8003	7573	30666	7382	6748	6542
	Rust	A	28119	21552	2.19378e+05	17951	22614	19018
		M	7258	6971	49213	6416	6186	6791
bitonic	C	A	7.59999e+06	2.37592e+06	7.39234e+06	2.37342e+06	2.38634e+06	2.38634e+06
		M	19950	10400	29151	11527	10374	10374
	Rust	A	1.25656e+07	8.58267e+06	5.17885e+07	7.56605e+06	1.66769e+06	1.61446e+06
		M	13091	15205	1.02875e+05	12825	14251	12357
bsort	C	A	2.79772e+05	3.16974e+05	8.66550e+05	1.88448e+05	3.17078e+05	3.17078e+05
		M	92884	68417	3.28215e+05	77994	68411	68411
	Rust	A	5.60760e+05	9.72825e+05	5.61152e+06	9.30995e+05	4.30652e+05	4.30652e+05
		M	98020	78495	1.45422e+06	89171	60530	60530
complex_updates	C	A	715	619	3062	596	634	669
		M	551	463	2484	457	473	532
	Rust	A	1785	1607	10263	1777	1599	1742
		M	635	570	5393	595	563	566
cosf	C	A	51706	52501	1.03783e+05	53406	54501	56207
		M	21183	19830	40422	19426	19885	19434
	Rust	A	33610	35270	1.37596e+05	32634	34642	34252
		M	15849	14846	60162	12189	12112	11853
cosf_libm	C	A	–	–	–	–	–	–
		M	–	–	–	–	–	–
	Rust	A	411266	545334	26803508	19357378	877615	876892
		M	232412	229108	281406	231382	225755	224610

count negative	C	A	5894	5409	28323	5396	4023	4188
		M	5467	4581	22240	4576	3591	3373
	Rust	A	13900	13011	60481	12871	17272	11683
		M	5488	4999	39834	4405	3548	3394
cubic	C	A	3.97878e+06	4.04081e+06	1.25715e+07	4.08217e+06	3.78503e+06	3.83597e+06
		M	1.08503e+06	1.05394e+06	2.75739e+06	1.05782e+06	9.33820e+05	9.38948e+05
	Rust	A	3.05065e+06	4.71865e+06	1.34994e+07	3.77641e+06	6.40034e+06	6.70035e+06
		M	9.77303e+05	8.29865e+05	3.72950e+06	9.07929e+05	1.01968e+06	8.84368e+05
cubic_libm	C	A	–	–	–	–	–	–
		M	–	–	–	–	–	–
	Rust	A	387955521	573997292	714479996	655435856	991564554	1044109336
		M	1374088	1145938	8143952	1894141	1109194	1108395
deg2rad	C	A	8366	8366	22803	8366	8263	8269
		M	9785	8355	20269	8348	8266	8281
	Rust	A	11981	11977	96081	39167	10003	10003
		M	11620	9789	70682	9544	9550	9550
fac	C	A	328	219	3079	364	292	292
		M	237	158	1030	234	209	199
	Rust	A	435	372	5397	956	649	649
		M	254	208	2132	332	295	296
fft	C	A	1.20939e+08	1.18522e+08	1.08023e+09	1.13738e+08	1.13766e+08	1.13766e+08
		M	4.67075e+05	4.44947e+05	1.43925e+06	3.99355e+05	3.18790e+05	3.18790e+05
	Rust	A	8.98447e+08	5.72214e+08	3.86030e+09	6.84928e+08	7.77635e+08	7.77629e+08
		M	1.30508e+06	5.16314e+05	5.70007e+06	4.46920e+05	4.74802e+05	4.74649e+05
filterbank	C	A	5.77051e+06	4.15094e+06	1.94598e+07	3.89062e+06	4.80743e+06	4.68212e+06
		M	5.19519e+06	3.41021e+06	1.57116e+07	3.18007e+06	2.01477e+06	2.01562e+06
	Rust	A	6.72939e+06	6.30142e+06	3.86152e+07	1.74242e+07	1.36339e+07	1.37031e+07
		M	5.73135e+06	3.45885e+06	2.35533e+07	2.10837e+06	2.15670e+06	2.15574e+06
fir2dim	C	A	7753	4736	22097	5644	1498	1707
		M	4597	3303	13244	3308	1192	1321
	Rust	A	42481	13088	4.14185e+05	12774	3821	3452
		M	5850	5335	1.35147e+05	6751	1328	1049
iir	C	A	284	227	824	217	226	217
		M	185	189	666	168	181	173
	Rust	A	493	473	4275	458	458	458
		M	207	182	2718	163	176	172
insertsort	C	A	1404	1222	7407	1268	2040	1217
		M	742	677	3325	660	663	636
	Rust	A	10194	5191	43332	7381	6701	6741
		M	1161	997	10993	1095	1022	1052
isqrt	C	A	6.25663e+05	4.84535e+05	2.67875e+06	4.53564e+05	3.38537e+05	3.42525e+05
		M	5.59593e+05	4.55508e+05	1.94162e+06	4.38507e+05	3.27464e+05	3.30449e+05
	Rust	A	5.60658e+05	4.24597e+05	8.21635e+06	2.91614e+05	2.89538e+05	2.89538e+05
		M	4.85554e+05	3.92459e+05	5.51586e+06	2.86421e+05	2.85400e+05	2.85400e+05

jfdctint	C	A	2612	2063	7253	1941	2096	2096
		M	2181	1763	5364	1703	1752	1754
	Rust	A	7049	6274	27668	9623	7132	7132
		M	2157	1818	14400	1885	1896	1896
lms	C	A	2.17430e+05	3.15014e+05	5.50371e+05	3.00996e+05	2.33482e+05	2.33485e+05
		M	1.80860e+05	2.54224e+05	4.60746e+05	2.63671e+05	1.85899e+05	1.85908e+05
	Rust	A	2.35344e+05	2.02741e+05	3.25325e+06	2.00099e+05	1.77739e+05	1.68336e+05
		M	1.99597e+05	1.59883e+05	1.92689e+06	1.24764e+05	1.14441e+05	1.07035e+05
ludcmp	C	A	1.79609e+05	1.75325e+05	1.81499e+05	1.65903e+05	2.48819e+05	2.46848e+05
		M	28181	27431	33933	27409	28738	29123
	Rust	A	5.05240e+05	7.52070e+05	7.30008e+05	1.00099e+06	4.68265e+05	4.49281e+05
		M	42591	35929	76943	38175	35879	35218
matrix1	C	A	21133	21478	1.27767e+05	21478	7528	7528
		M	11472	9922	45756	9922	5078	5078
	Rust	A	56696	54946	3.10533e+05	38378	35451	12337
		M	10806	9350	1.83323e+05	6668	5192	5006
md5	C	A	1.59773e+07	1.00793e+07	4.82222e+07	9.85904e+06	9.35740e+06	8.05782e+06
		M	1.39078e+07	7.51179e+06	3.91140e+07	7.41800e+06	6.79560e+06	5.58678e+06
	Rust	A	2.67802e+08	1.00902e+08	3.62737e+08	2.76790e+08	1.03913e+08	1.02905e+08
		M	1.18817e+07	7.53376e+06	1.07960e+08	8.98743e+06	7.75098e+06	7.39343e+06
minver	C	A	65149	60861	71846	60647	59232	60404
		M	15855	15305	20872	15495	15962	16085
	Rust	A	1.05430e+05	1.23136e+05	2.33549e+05	1.14687e+05	1.14689e+05	1.14909e+05
		M	18596	18236	39147	17942	17942	17610
pm	C	A	–	1.01265e+07	5.06308e+07	2.23741e+07	1.53740e+07	1.53031e+07
		M	9.68256e+06	8.09053e+06	3.03834e+07	8.28197e+06	6.95550e+06	6.92455e+06
	Rust	A	83818797	1.27844e+07	1.87793e+08	2.93548e+07	5.01843e+07	2.29828e+07
		M	7.18486e+06	6.87027e+06	1.10340e+08	8.07012e+06	5.20546e+06	5.05682e+06
pm_libm	C	A	–	–	–	–	–	–
		M	–	–	–	–	–	–
	Rust	A	100232980	13788670	194337446	10420991	13157789	33209939
		M	6901485	6986954	95643688	8289998	5523320	5257850
prime	C	A	798	795	2531	809	885	903
		M	293	284	1068	285	256	284
	Rust	A	952	1080	17585	3053	1090	1021
		M	368	416	6955	301	287	288
quicksort	C	A	–	7.67035e+09	2.94751e+10	1.89716e+10	1.89856e+10	1.97428e+10
		M	3.50971e+06	3.20466e+06	8.69065e+06	3.33997e+06	3.19234e+06	3.20832e+06
	Rust	A	6740223740	9.58762e+09	1.34946e+10	9.24174e+09	9.62108e+09	9.40357e+09
		M	2.07099e+06	1.60585e+06	5.00550e+06	1.58337e+06	1.62860e+06	1.62812e+06
quicksort_libm	C	A	–	–	–	–	–	–
		M	–	–	–	–	–	–
	Rust	A	6736497508	9317197781	13665390755	9090493142	9416611413	9359672523
		M	2071338	1609169	5052569	1586475	1631132	1631113

rad2deg	C	A	9432	8351	23100	8351	8266	8272
		M	9770	8334	20573	8331	8242	8255
	Rust	A	9466	8371	42215	32227	6924	6936
		M	9438	8343	21461	8257	6812	6816
recursion	C	A	14515	11670	18760	11694	11671	10795
		M	2694	2028	7762	2587	2035	2028
	Rust	A	14550	10835	18726	11041	10842	10833
		M	2705	2042	6922	2602	2057	2056
sha	C	A	4.58285e+06	3.09095e+06	9.84881e+06	3.12629e+06	1.61419e+06	1.64373e+06
		M	2.42477e+06	1.90217e+06	5.77597e+06	1.92111e+06	1.19647e+06	1.19393e+06
	Rust	A	7.08876e+06	5.61615e+06	4.86647e+07	3.13081e+06	2.90862e+06	2.88930e+06
		M	2.32663e+06	1.78577e+06	2.14519e+07	1.39987e+06	1.10027e+06	1.12856e+06
st	C	A	75996	66643	3.40342e+05	63556	43828	57055
		M	68664	64222	2.76689e+05	57076	37082	38556
	Rust	A	88849	94166	1.07796e+06	5.07104e+05	99933	83016
		M	51381	58146	7.36052e+05	39644	35707	36952
st_libm	C	A	–	–	–	–	–	–
		M	–	–	–	–	–	–
	Rust	A	225072	128418	47560852	5584793	185839	181432
		M	142539	95929	1384042	406577	76291	76322

Table 2: Benchmark Cycle Counts (sequential)

			Oz	Os	O0	O1	O2	O3
adpcm_dec	C	A	–	–	9506	–	–	3640
		M	2501	3070	6809	2999	2772	2583
	Rust	A	–	–	82990	–	–	14667
		M	5900	2874	42987	3506	2429	2443
adpcm_enc	C	A	–	–	14256	–	–	5682
		M	3499	3800	9503	3740	3997	3890
	Rust	A	–	–	1.17410e+05	–	–	18767
		M	3616	3161	37695	3002	2694	2717
ammunition	C	A	–	–	3.98503e+09	–	–	7.78934e+08
		M	2.86698e+08	2.63608e+08	5.25244e+08	1.95519e+08	2.12395e+08	2.15352e+08
	Rust	A	–	–	1.04218e+10	–	–	4.05914e+09
		M	3.31515e+08	2.83844e+08	1.54582e+09	2.17711e+08	1.62134e+08	1.61043e+08
anagram	C	A	–	–	216252206	–	–	1.56412e+08
		M	2.00403e+06	1.63583e+06	7.93381e+06	1.64987e+06	1.62347e+06	1.62586e+06
	Rust	A	–	–	–	–	–	1.65462e+11
		M	9.22187e+06	2.22067e+06	2.17255e+07	1.58577e+06	1.59284e+06	1.59799e+06
audiobeam	C	A	–	–	2.04689e+06	–	–	3.22477e+06
		M	4.45473e+05	4.30607e+05	1.11869e+06	3.71615e+05	3.81926e+05	3.37495e+05
	Rust	A	–	–	3.26688e+07	–	–	1.18446e+06
		M	2.20110e+06	7.89519e+05	9.40187e+06	1.55942e+06	7.95530e+05	5.13939e+05
cjpeg_transupp	C	A	–	–	5.52758e+07	–	–	3.15137e+06
		M	2.45251e+06	2.11204e+06	9.05935e+06	2.26736e+06	9.00100e+05	6.19404e+05
	Rust	A	–	–	1.63835e+09	–	–	2.68455e+07
		M	4.37458e+06	3.97047e+06	4.94271e+07	2.25565e+06	2.32222e+06	1.14652e+06
cjpeg_wrbmp	C	A	–	–	4.20487e+05	–	–	1.08616e+05
		M	54245	47799	2.84747e+05	53625	43934	43409
	Rust	A	–	–	1.45944e+06	–	–	3.42702e+05
		M	3.10213e+05	47728	5.48945e+05	60762	47199	47478
dijkstra	C	A	–	–	1.32294e+11	–	–	5.24683e+10
		M	3.57241e+07	3.17990e+07	1.29879e+08	3.14676e+07	3.53358e+07	2.77661e+07
	Rust	A	–	–	3.74174e+11	–	–	1.15082e+11
		M	4.96402e+07	4.17254e+07	4.69603e+08	5.79379e+07	4.61372e+07	4.31723e+07
epic	C	A	–	–	2.09990e+09	–	–	77446
		M	3.81356e+06	3.66187e+06	1.31282e+07	3.99439e+06	5.90104e+06	5.60404e+06
	Rust	A	–	–	3.27428e+09	–	–	1.02446e+09
		M	5.33536e+06	5.44895e+06	1.96293e+07	5.73625e+06	5.43146e+06	5.72935e+06

fmref	C	A	–	–	6.75594e+06	–	–	4.01093e+06
		M	2.02612e+06	1.97141e+06	2.61743e+06	2.03719e+06	2.02710e+06	2.04246e+06
	Rust	A	–	–	8.26285e+07	–	–	1.53155e+07
		M	1.21608e+06	1.18134e+06	3.33299e+06	1.20900e+06	1.12492e+06	1.09221e+06
g723_enc	C	A	–	–	2.22252e+06	–	–	5.30026e+05
		M	6.79933e+05	5.48080e+05	1.42494e+06	4.97028e+05	3.18187e+05	3.07986e+05
	Rust	A	–	–	1.56455e+07	–	–	5.08547e+05
		M	6.97238e+05	5.32994e+05	4.75449e+06	2.60907e+05	2.53120e+05	2.49506e+05
gsm_dec	C	A	–	–	2.21263e+07	–	–	2.08785e+08
		M	1.28891e+06	1.20664e+06	4.21280e+06	1.21958e+06	1.13894e+06	1.01777e+06
	Rust	A	–	–	8.52888e+07	–	–	2.16896e+06
		M	1.24802e+06	1.11185e+06	1.58361e+07	1.22139e+06	8.43732e+05	8.61035e+05
gsm_enc	C	A	–	–	3.08450e+07	–	–	1.68224e+07
		M	3.21160e+06	2.78451e+06	9.20965e+06	2.96993e+06	2.87397e+06	2.49396e+06
	Rust	A	–	–	2.86521e+08	–	–	2.86809e+07
		M	1.71501e+07	6.74687e+06	1.18043e+08	6.94327e+06	6.40299e+06	6.37484e+06
h264_dec	C	A	–	–	1.64818e+05	–	–	56738
		M	31298	28713	89365	30260	27022	24294
	Rust	A	–	–	1.56575e+06	–	–	2.73378e+05
		M	28062	23540	1.43933e+05	22960	23002	22112
huff_dec	C	A	–	–	2.15262e+06	–	–	2.15008e+06
		M	1.87187e+05	1.46803e+05	5.39221e+05	1.11663e+05	1.08517e+05	1.00009e+05
	Rust	A	–	–	1.68456e+07	–	–	1.48355e+06
		M	3.00207e+05	2.05084e+05	2.18683e+06	1.40871e+05	1.49499e+05	1.47629e+05
huff_enc	C	A	–	–	5.42244e+10	–	–	2.95925e+10
		M	6.09715e+05	4.93575e+05	1.68008e+06	4.32435e+05	3.94416e+05	3.85478e+05
	Rust	A	–	–	1.07317e+11	–	–	1.30023e+11
		M	7.28927e+05	5.22275e+05	6.54745e+06	5.49000e+05	4.92898e+05	4.11944e+05
mpeg2	C	A	–	–	3.59834e+09	–	–	1.06997e+09
		M	6.25249e+07	6.38131e+07	1.51334e+08	6.32670e+07	5.79999e+07	5.37841e+07
	Rust	A	–	–	9.85356e+10	–	–	2.64341e+09
		M	1.01795e+08	1.06816e+08	8.17998e+08	9.84650e+07	9.42451e+07	1.06059e+08
ndes	C	A	–	–	1.79151e+05	–	–	56497
		M	46840	46789	1.38732e+05	47190	44045	44143
	Rust	A	–	–	9.10101e+05	–	–	2.71335e+05
		M	1.46166e+05	1.05650e+05	3.43898e+05	72658	72121	72178
petrinet	C	A	–	–	8827	–	–	4212
		M	1359	1633	3750	1641	1642	1572
	Rust	A	–	–	1.43515e+05	–	–	36575
		M	4083	3982	39815	4708	3991	3631

rijndael_dec	C	A	–	–	1.55862e+07	–	–	9.55458e+06
		M	6.90043e+06	6.58626e+06	9.29011e+06	6.66000e+06	6.66191e+06	6.66810e+06
	Rust	A	–	–	6.53439e+07	–	–	9.05265e+06
		M	7.49503e+06	6.44099e+06	2.17959e+07	6.98656e+06	6.14469e+06	6.37667e+06
rijndael_enc	C	A	–	–	1.25335e+07	–	–	9.43318e+06
		M	6.70240e+06	6.53307e+06	9.00073e+06	6.62877e+06	6.91690e+06	6.79954e+06
	Rust	A	–	–	6.59143e+07	–	–	8.65989e+06
		M	7.25047e+06	6.29557e+06	2.21509e+07	6.66280e+06	6.14664e+06	6.40517e+06
statemate	C	A	–	–	2.91638e+05	–	–	1.18304e+05
		M	38566	74202	1.03459e+05	69266	66016	32969
	Rust	A	–	–	1.31728e+06	–	–	4.82076e+05
		M	41674	42770	2.80812e+05	44228	42853	41437
susan	C	A	–	–	5.51861e+08	–	–	1.33833e+08
		M	3.17106e+07	2.79250e+07	1.19489e+08	3.48480e+07	2.75411e+07	2.75630e+07
	Rust	A	–	–	6.21746e+09	–	–	7.01204e+09
		M	2.56517e+08	2.33517e+08	1.40821e+09	3.10607e+08	2.08803e+08	2.08377e+08

Table 3: Code Size (kernel)

		Oz	Os	O0	O1	O2	O3
binarysearch	C	2624	2636	2784	2636	2880	3412
	Rust	2140	2160	8248	3752	2260	2260
bitcount	C	3560	3740	4232	3688	3892	4096
	Rust	3584	3704	12936	5292	4444	13096
bitonic	C	2684	2664	2864	2644	2796	2796
	Rust	2296	2352	8360	3868	2688	2772
bsort	C	2624	2632	2792	2592	2772	2776
	Rust	2100	2108	10036	2712	3216	3872
complex_updates	C	2816	3040	3120	3036	3820	4300
	Rust	2308	2496	9572	3196	2768	3548
cosf	C	3572	3772	4688	4280	4280	4332
	Rust	2940	3020	10048	4900	3456	3476
cosf_libm	C	–	–	–	–	–	–
	Rust	8812	8876	26652	11064	11280	11424
countnegative	C	2672	2720	2932	2716	3152	3680
	Rust	2120	2144	6220	2788	2720	2892
cubic	C	6604	7392	10808	8204	8648	8472
	Rust	5620	6072	17188	9920	9396	11752
cubic_libm	C	–	–	–	–	–	–
	Rust	11596	11876	37508	15160	14152	14264
deg2rad	C	2560	2568	2676	2568	2648	2632
	Rust	2040	2036	6056	2156	2140	2140
fac	C	2512	2528	2632	2544	2600	2584
	Rust	2028	2044	6132	2064	2288	2288
fft	C	15388	15344	15996	15320	16784	16784
	Rust	15876	15640	25768	18084	16804	16840
filterbank	C	2992	3216	3412	3252	5504	5608
	Rust	3544	2748	12404	6340	8016	8028
fir2dim	C	3056	3188	3608	3376	5492	7288
	Rust	2784	2996	11816	3944	3960	5120
iir	C	2696	2736	3012	2724	3504	3512
	Rust	2240	2436	10300	2500	2500	2856
insertsort	C	2732	2816	3100	2816	3044	3456
	Rust	2536	2576	9000	4160	2708	3036
isqrt	C	2612	2656	2864	2704	3104	3080
	Rust	2108	2124	10224	2496	2472	2472

jfdctint	C	3120	3164	3624	3176	4164	4320
	Rust	2796	2868	12432	5136	4524	4528
lms	C	5856	6296	6632	6212	7476	7476
	Rust	7652	7852	20268	10060	9012	9044
ludcmp	C	5836	5992	6596	6036	8060	8980
	Rust	6796	6732	16600	8996	7172	7852
matrix1	C	2664	2844	2860	2840	3400	3400
	Rust	2204	2388	8448	2752	3008	3948
md5	C	5116	5240	7032	5444	6468	7280
	Rust	5636	5456	22748	7964	6564	6620
minver	C	6280	6200	7060	6192	9600	9704
	Rust	9348	9340	19636	12764	11404	11412
pm	C	36248	37084	39196	36992	41420	41612
	Rust	37568	37624	58112	43964	45084	47976
pm_libm	C	–	–	–	–	–	–
	Rust	38216	37892	66556	46492	44884	52488
prime	C	2608	2700	2864	2708	2820	2804
	Rust	2124	2144	6652	2324	2308	2308
quicksort	C	23364	23696	26048	23684	24928	25116
	Rust	30916	30992	43928	32932	31272	31636
quicksort_libm	C	–	–	–	–	–	–
	Rust	33324	33284	58288	40532	33412	33444
rad2deg	C	2564	2572	2680	2572	2652	2636
	Rust	2100	2104	2248	2184	2152	2144
recursion	C	2504	2532	2608	2532	2536	2520
	Rust	2040	2060	2188	2056	2060	2056
sha	C	36456	36744	37492	36772	39424	39500
	Rust	36712	36740	44900	37712	37888	39264
st	C	4528	4680	5100	4924	7916	7796
	Rust	2872	3072	8972	4880	6572	6476
st_libm	C	–	–	–	–	–	–
	Rust	3380	3564	21836	5968	5788	5804

Table 4: Code Size (sequential)

		Oz	Os	O0	O1	O2	O3
adpcm_dec	C	4528	5544	6132	5596	6036	5468
	Rust	5408	4540	17480	7892	5716	6108
adpcm_enc	C	4728	5652	6460	5700	6744	6760
	Rust	4440	6172	16516	7372	6044	6020
ammunition	C	17436	20420	27792	22128	38776	40248
	Rust	23236	25092	60172	48944	47460	56904
anagram	C	24872	25320	26372	25444	27280	27696
	Rust	37036	35840	48156	37052	35856	35988
audiobeam	C	31008	31572	33676	31968	35464	36540
	Rust	34988	33528	58720	39748	35572	37764
cjpeg_transupp	C	4384	4788	5924	4996	11140	14916
	Rust	4768	4864	16668	14420	13376	21232
cjpeg_wrbmp	C	3820	3992	4012	4064	6472	6508
	Rust	4516	5380	16600	6992	5376	5408
dijkstra	C	12936	13020	13340	13096	13576	13940
	Rust	12856	12888	21628	15088	13552	13772
epic	C	21552	22036	23140	22924	26784	30364
	Rust	22692	23284	33924	25564	24456	25476
fmref	C	13956	14636	16292	16508	19132	18972
	Rust	10652	10828	26896	14448	13224	13880
g723_enc	C	5420	5944	7188	6208	7536	8036
	Rust	5088	6640	14528	8532	7152	7172
gsm_dec	C	12580	12740	16096	13512	16368	16540
	Rust	13136	14364	28276	17968	17288	18988
gsm_enc	C	16308	16424	24464	17148	24540	25656
	Rust	16572	14556	305704	21564	19536	22636
h264_dec	C	11396	11464	17104	11476	12800	13792
	Rust	11868	12052	23124	14564	12588	12960
huff_dec	C	4476	4560	5252	5680	7444	7292
	Rust	4760	4836	14904	8212	7748	7908
huff_enc	C	5228	5324	7008	6396	8676	8424
	Rust	7136	6264	18476	8660	8512	11016
mpeg2	C	34600	35240	40692	38248	46248	90452
	Rust	32588	32400	44696	35308	33124	37792
ndes	C	6652	6784	7492	6836	7852	9292
	Rust	10300	10692	20156	12516	11076	11108

petrinet	C	3704	4552	7052	4556	4560	4388
	Rust	3496	3560	12356	5224	3660	3992
rijndael_dec	C	55980	56424	57376	56448	57016	57064
	Rust	57744	56880	87968	67736	58572	60272
rijndael_enc	C	50792	51528	52216	51616	52364	52676
	Rust	53068	51984	83368	63112	53796	55464
statemate	C	5196	8220	10748	8360	8336	5844
	Rust	8012	8412	19676	10428	9232	9080
susan	C	24968	26140	36988	26844	31592	32504
	Rust	32748	34112	78144	49476	38316	40848

B) C `Og` data

Table 5: C `Og` Kernel

Benchmark	Measured	WCET Analysis	Size
`binarysearch`	97	121	2636
`bitcount`	7382	24328	3688
`bitonic`	11527	2373419	2644
`bsort`	77994	188448	2592
`complex_updates`	457	596	3036
`cosf`	19426	53406	4280
`countnegative`	4576	5396	2716
`cubic`	1057819	4082168	8204
`deg2rad`	8348	8366	2568
`fac`	234	364	2544
`fft`	399355	113738001	15320
`filterbank`	3180069	3890617	3252
`fir2dim`	3308	5644	3376
`iir`	168	217	2724
`insertsort`	660	1268	2816
`isqrt`	438507	453564	2704
`jfdctint`	1703	1941	3176
`lms`	263671	300996	6212
`ludcmp`	27409	165903	6036
`matrix1`	9922	21478	2840
`md5`	7417993	9859808	5444
`minver`	15495	60647	6192
`pm`	8281978	22374094	36992
`prime`	285	809	2708
`quicksort`	3339971	18971637819	23684
`rad2deg`	8331	8351	2572
`recursion`	2587	11694	2532
`sha`	1921107	3126289	36772
`st`	57076	63556	4924

Table 6: C `Og` Sequential

Benchmark	Measured	Size
adpcm_dec	2999	5596
adpcm_enc	3740	5700
ammunition	195519394	22128
anagram	1649868	25444
audiobeam	371615	31968
cjpeg_transupp	2267365	4996
cjpeg_wrbmp	53625	4064
dijkstra	31467565	13096
epic	3994393	22924
fmref	2037187	16508
g723_enc	497028	6208
gsm_dec	1219583	13512
gsm_enc	2969939	17148
h264_dec	30260	11476
huff_dec	111663	5680
huff_enc	432435	6396
mpeg2	63266958	38248
ndes	47190	6836
petrinet	1641	4556
rijndael_dec	6660021	56448
rijndael_enc	6628769	51616
statemate	69266	8360
susan	34848000	26844

C) 2. Run Measurements

Table 7: C kernel warm-run

Benchmark	O0	O1	O2	O3	Og	Os	Oz
binarysearch	273	85	77	78	85	73	122
bitcount	30524	7312	6659	6441	7312	7482	7947
bitonic	29119	11505	10341	10341	11505	10372	19934
bsort	328185	77985	68396	68396	77985	68397	92869
complex_updates	2466	434	400	385	434	451	528
cosf	40121	19099	19570	19117	19099	19608	21012
countnegative	22195	4565	3578	3320	4565	4569	5458
cubic	2.76e+06	1.06e+06	933480	938444	1.06e+06	1.05e+06	1.08e+06
deg2rad	20245	8326	8241	8243	8326	8326	9771
fac	1022	222	192	182	222	158	227
fft	1.44e+06	399321	318695	318695	399321	444908	467075
filterbank	1.57e+07	3.18e+06	2.01e+06	2.02e+06	3.18e+06	3.41e+06	5.20e+06
fir2dim	13087	3222	1084	1146	3222	3251	4514
iir	640	145	135	136	145	171	173
insertsort	3268	631	597	526	631	654	725
isqrt	1.94e+06	438481	327369	330364	438481	455489	559583
jfdctint	5300	1604	1612	1612	1604	1666	2104
lms	460746	263632	185860	185868	263641	254163	180827
ludcmp	33661	27124	28422	28930	27124	27191	27978
matrix1	45731	9910	5046	5046	9910	9910	11454
md5	3.91e+07	7.42e+06	6.80e+06	5.59e+06	7.42e+06	7.51e+06	1.39e+07
minver	20392	15025	15811	15868	15025	14874	15418
pm	3.04e+07	8.28e+06	6.96e+06	6.92e+06	8.28e+06	8.09e+06	9.68e+06
prime	1027	259	241	248	259	259	293
quicksort	8.69e+06	3.34e+06	3.19e+06	3.21e+06	3.34e+06	3.20e+06	3.51e+06
rad2deg	20549	8313	8214	8219	8313	8313	9744
recursion	7752	2581	2024	2024	2581	2024	2689
sha	5.78e+06	1.92e+06	1.20e+06	1.19e+06	1.92e+06	1.90e+06	2.42e+06
st	276538	56965	36915	38370	56965	64108	68591

Table 8: Rust kernel warm-run

Benchmark	O0	O1	O2	O3	Os	Oz
binarysearch	544	129	125	125	127	134
bitcount	49067	6226	5877	6727	6787	7125
bitonic	102783	12799	14224	12316	15170	13063
bsort	1.45e+06	89118	60479	60479	78483	98009
complex_updates	5283	532	509	416	546	623
cosf	59805	11945	12	15	14680	15705
cosf_libm	280998	230831	225262	224119	228634	232003
countnegative	39770	4361	3518	3364	4982	5478
cubic	3.73e+06	907480	1.02e+06	884282	829238	976782
cubic_libm	8.14e+06	1.89e+06	1.11e+06	1.11e+06	1.15e+06	1.37e+06
deg2rad	70614	11	11	11	9785	11587
fac	2069	314	236	239	198	236
fft	5.70e+06	446756	474665	474484	516203	1.30e+06
filterbank	2.36e+07	2.11e+06	2.16e+06	2.16e+06	3.46e+06	5.73e+06
fir2dim	134875	6648	1185	858	5247	5789
iir	2485	123	127	124	165	193
insertsort	10855	1032	961	981	956	1129
isqrt	5.52e+06	286350	285337	285337	392440	485539
jfdctint	14094	1748	1739	1741	1719	2076
lms	1.93e+06	124683	114402	106927	159853	199563
ludcmp	76849	37825	35540	34896	35606	42351
matrix1	183256	6634	5147	4975	9329	10781
md5	1.08e+08	8.99e+06	7.75e+06	7.39e+06	7.53e+06	1.19e+07
minver		17792	17770	17410	17913	18268
pm	1.10e+08	8.07e+06	5.21e+06	5.06e+06	6.87e+06	7.18e+06
pm_libm	9.56e+07	8.29e+06	5.52e+06	5.26e+06	6.99e+06	6.90e+06
prime	6829	266	249	255	382	341
quicksort	5.01e+06	1.58e+06	1.63e+06	1.63e+06	1.61e+06	2.07e+06
quicksort_libm	5.05e+06	1.59e+06	1.63e+06	1.63e+06	1.61e+06	2.07e+06
rad2deg	21431	8232	6793	6810	8335	9410
recursion	6898	2592	2036	2036	2031	2696
sha	2.15e+07	1.40e+06	1.10e+06	1.13e+06	1.79e+06	2.33e+06
st	735824	39275	35268	36659	58049	51323
st_libm	1.38e+06	406304	75882	75933	95776	142427

Table 9: C sequential warm-run

Benchmark	O0	O1	O2	O3	Og	Os	Oz
adpcm_dec	6577	2792	2519	2367	2792	2876	2347
adpcm_enc	9233	3511	3745	3572	3511	3566	3333
ammunition	5.25e+08	1.96e+08	2.12e+08	2.15e+08	1.96e+08	2.64e+08	2.87e+08
anagram	7.93e+06	1.65e+06	1.62e+06	1.63e+06	1.65e+06	1.64e+06	2.00e+06
audiobeam	1.12e+06	371266	381867	337480	371247	430344	445058
cjpeg_transupp	9.06e+06	2.27e+06	900086	619399	2.27e+06	2.11e+06	2.45e+06
cjpeg_wrbmp	284643	53530	43658	43195	53530	47710	54172
dijkstra	1.30e+08	3.15e+07	3.53e+07	2.78e+07	3.15e+07	3.18e+07	3.57e+07
epic	1.31e+07	3.99e+06	5.90e+06	5.60e+06	3.99e+06	3.66e+06	3.81e+06
fmref	2.54e+06	2.02e+06	2.01e+06	2.03e+06	2.02e+06	1.95e+06	2.00e+06
g723_enc	1.42e+06	496650	317735	307592	496650	547802	679752
gsm_dec	4.21e+06	1.22e+06	1.14e+06	1.02e+06	1.22e+06	1.21e+06	1.29e+06
gsm_enc	9.21e+06	2.97e+06	2.87e+06	2.49e+06	2.97e+06	2.78e+06	3.21e+06
h264_dec	89184	30185	26925	24106	30185	28631	31232
huff_dec	539019	111399	108127	99597	111399	146666	187084
huff_enc	1.68e+06	432092	394170	385125	432092	493384	609463
mpeg2	1.51e+08	6.33e+07	5.80e+07	5.38e+07	6.33e+07	6.38e+07	6.25e+07
ndes	137522	46836	43649	43649	46836	46356	46388
petrinet	3285	1261	1267	1218	1261	1260	1226
rijndael_dec	9.29e+06	6.66e+06	6.66e+06	6.67e+06	6.66e+06	6.59e+06	6.90e+06
rijndael_enc	9.00e+06	6.63e+06	6.92e+06	6.80e+06	6.63e+06	6.53e+06	6.70e+06
statemate	103195	68867	65666	32749	68867	73872	38358
susan	1.19e+08	3.48e+07	2.75e+07	2.76e+07	3.48e+07	2.79e+07	3.17e+07

Table 10: Rust sequential warm-run

Benchmark	O0	O1	O2	O3	Os	Oz
adpcm_dec	42920	3224	2141	2201	2666	5599
adpcm_enc	37582	2691	2290	2376	2861	3402
ammunition	1.55e+09	2.18e+08	1.62e+08	1.61e+08	2.84e+08	3.32e+08
anagram	2.17e+07	1.59e+06	1.59e+06	1.60e+06	2.22e+06	9.22e+06
audiobeam	9.40e+06	1.56e+06	795491	513918	789326	2.20e+06
cjpeg_transupp	4.94e+07	2.26e+06	2.32e+06	1.15e+06	3.97e+06	4.37e+06
cjpeg_wrbmp	548945	60625	47117	47373	47661	310150
dijkstra	4.70e+08	5.79e+07	4.61e+07	4.32e+07	4.17e+07	4.96e+07
epic	1.96e+07	5.74e+06	5.43e+06	5.73e+06	5.45e+06	5.34e+06
fmref	3.33e+06	1.21e+06	1.12e+06	1.09e+06	1.18e+06	1.22e+06
g723_enc	4.75e+06	260453	252657	249071	532708	696954
gsm_dec	1.58e+07	1.22e+06	843585	860855	1.11e+06	1.25e+06
gsm_enc		6.94e+06	6.40e+06	6.37e+06	6.75e+06	1.71e+07
h264_dec	143911	22749	22789	21879	23358	27924
huff_dec	2.19e+06	140488	149110	147262	204870	300016
huff_enc	6.55e+06	548544	492412	411713	521845	728496
mpeg2	8.18e+08	9.85e+07	9.42e+07	1.06e+08	1.07e+08	1.02e+08
ndes	343799	72384	71870	71813	105353	145812
petrinet	39531	4610	3888	3472	3892	3993
rijndael_dec	2.18e+07	6.99e+06	6.14e+06	6.38e+06	6.44e+06	7.49e+06
rijndael_enc	2.22e+07	6.66e+06	6.15e+06	6.40e+06	6.30e+06	7.25e+06
statemate	280666	44051	42696	41320	42605	41598
susan	1.41e+09	3.11e+08	2.09e+08	2.08e+08	2.34e+08	2.57e+08